RAJALAKSHMI ENGINEERING COLLEGE

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D2

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int n;
    scanf("%d", &n);

    int count = 0;

    for (int i = 1; i <= n; ++i)
    {
        count++;   // if condition

        if (n % i == 0)
        {
            count++;   // printf
        }
    }

    count += n + 1;  // for-loop condition checks

    printf("%d", count);

    return 0;
}
```

Check

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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